LEARNING SESSION S1

Advanced Excel: Basic to Advanced

Every Excel skill an analyst needs, from the first click to Power Query, dynamic arrays, VBA and Copilot

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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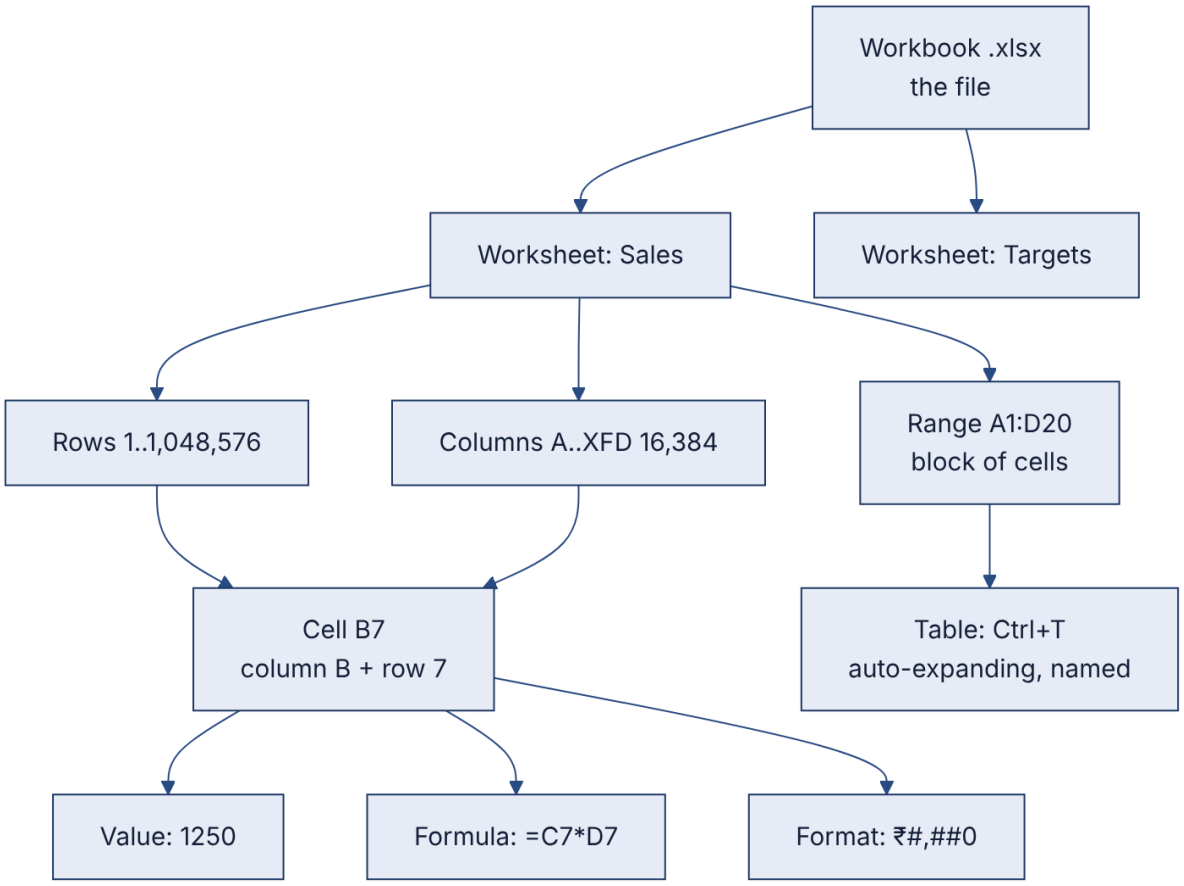
Learning session · non-tech friendly · companion workbook: Excel_Basic_to_Advanced.xlsx (one sheet per topic, live formulas)

Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#s1**

Roadmap

This session takes you from opening Excel for the first time to building automated dashboards. Each level builds on the previous one. Every topic has a matching sheet in the **Excel Basic-to-Advanced workbook** (see Downloads) with live examples you can edit.

Level	You will learn	Workbook sheets
1 · Foundations	Interface, data entry, cell references, formatting, number formats	01 Environment, 02 References, 03 Custom Formats
2 · Core functions	Math, text, date/time, aggregate and conditional aggregates	04 Number, 05 Text, 06 Date & Time, 07 Aggregate
3 · Logic & lookups	IF family, error handling, VLOOKUP, INDEX/MATCH, XLOOKUP	08 Logical, 09 Lookup
4 · Data tools	Sort, filter, tables, validation, conditional formatting, cleaning	10 Conditional Formatting, 11 Validation, 12 Sort & Filter
5 · Analysis	Pivot tables, slicers, what-if analysis, statistics	13 Pivot Data, 14 What-If
6 · Modern Excel	Dynamic arrays, LET, LAMBDA, GROUPBY, REGEX	15 Dynamic Arrays
7 · Power tools	Power Query, Power Pivot, DAX, data model	16 Power Query Steps
8 · Charts & dashboards	Chart choice, combo/Pareto/waterfall, sparklines, dashboard design	17 Charts, 18 Dashboard Case Study
9 · Automation & AI	Macros/VBA, Office Scripts, Power Automate, Copilot, Python in Excel	19 Automation & AI



The building blocks of every workbook.

Tip Learn by doing: open the workbook, change the blue input cells, and watch the black formula cells update. That feedback loop is how Excel skill sticks.

Level 1 · Foundations

Getting around

- Enter data: type and press `Enter` (down) or `Tab` (right). `Ctrl` + `Enter` fills the same value into all selected cells.
- Edit: `F2`; undo `Ctrl` + `Z`; repeat `F4`.
- Series: type 1, 2 and drag; or type Jan and drag; double-click the fill handle to fill down to the end of adjacent data.
- Flash Fill `Ctrl` + `E`: type the first answer by hand; Excel copies the pattern.

How Excel stores values

Numbers	Right-aligned by default. Up to 15 significant digits (longer IDs like card numbers must be stored as text).

Text	Left-aligned. Numbers stored as text show a green triangle; fix with "Convert to Number" or `=VALUE()`.
Dates	Serial numbers: 1 = 1 Jan 1900; 1 Oct 2026 = 46296. Times are fractions of a day (0.5 = noon).
Booleans / errors	TRUE/FALSE; errors such as #N/A, #VALUE!, #DIV/0!, #REF!, #NAME?, #SPILL!

Cell references

Reference	Copy down	Copy right	Typical use
A1	A2, A3…	B1, C1…	Row-by-row calculations
$A$1	stays	stays	A fixed rate (GST, exchange rate)
A$1	stays	B$1…	Header row in a grid
$A1	$A2…	stays	First column in a grid
Sheet2!B5 / 'Sales Data'!B5	Another sheet (quote names with spaces)		Link inputs
Sales[Amount]	Structured reference to a Table column		Auto-expanding formulas
Named range GST_Rate	Formulas → Define Name		Readable formulas: `=Amount*GST_Rate`

Formatting essentials

- Format Cells `Ctrl` + `1`: Number, Alignment (wrap, merge vs **Center Across Selection**, which is safer), Font, Border, Fill, Protection.
- Cell Styles and Format Painter (double-click to paint many times).
- Custom number formats: see the table in **Building Blocks · Class 1**, including the Indian lakh format and ▲▼ indicators.

Level 2 · Core functions

The 40 functions that cover most analyst work

Group	Functions	Real example
Math	SUM, PRODUCT, ROUND/ROUNDUP/ROUNDDOWN, INT, MOD, ABS, POWER, SQRT, CEILING.MATH, FLOOR.MATH, RAND, RANDBETWEEN, SUMPRODUCT	`=SUMPRODUCT(Qty,Price)` total sales without a helper column
Text	LEFT, RIGHT, MID, LEN, FIND, SEARCH, SUBSTITUTE, REPLACE, TRIM, CLEAN, UPPER, LOWER, PROPER, TEXT, VALUE, CONCAT, TEXTJOIN, EXACT, REPT	`=PROPER(TRIM(A2))` tidy imported names

Group	Functions	Real example
Date & time	TODAY, NOW, DATE, YEAR, MONTH, DAY, EOMONTH, EDATE, WEEKDAY, WEEKNUM, NETWORKDAYS(.INTL), WORKDAY, DATEDIF, YEARFRAC, HOUR, MINUTE	`=EOMONTH(TODAY(),-1)+1` first day of this month
Aggregate	COUNT, COUNTA, COUNTBLANK, AVERAGE, MEDIAN, MODE, MIN, MAX, SMALL, LARGE, RANK.EQ, PERCENTILE, QUARTILE, STDEV.S	`=RANK.EQ(H2,$H$2:$H$100)` sales-rep ranking
Conditional aggregate	SUMIF(S), COUNTIF(S), AVERAGEIF(S), MAXIFS, MINIFS	`=MAXIFS(Amount,City,"Pune")` biggest Pune order
Visible-rows aggregate	SUBTOTAL, AGGREGATE	`=SUBTOTAL(109,H:H)` sum of filtered rows

Criteria tricks in SUMIFS / COUNTIFS

```
=SUMIFS(Amount, City, "Pune")                      exact match
=SUMIFS(Amount, Product, "*phone*")                wildcard: contains "phone"
=SUMIFS(Amount, Amount, ">=10000")                 comparison as text
=SUMIFS(Amount, Date, ">="&$K$1, Date, "<="&$K$2)  between two dates held in cells
=SUM(SUMIFS(Amount, City, {"Pune","Mumbai"}))        OR across two cities
=COUNTIFS(Region, "<>North")                       not equal
=COUNTIF(A:A, A2)>1                                duplicate check
```

Text recipes

```
First name              =LEFT(A2, FIND(" ", A2) - 1)
Last name               =RIGHT(A2, LEN(A2) - FIND(" ", A2))
Email domain            =MID(B2, FIND("@", B2) + 1, 100)
Mask a phone number     ="XXXXXX" & RIGHT(C2, 4)
Count words             =LEN(TRIM(A2)) - LEN(SUBSTITUTE(A2, " ", "")) + 1
Pad an ID to 6 digits   =TEXT(D2, "000000")
```

Date recipes

```
Age in years            =DATEDIF(DOB, TODAY(), "Y")
Financial year (Apr–Mar) ="FY" & IF(MONTH(B2)>=4, YEAR(B2)+1, YEAR(B2))
Quarter (Apr start)     ="Q" & ROUNDUP(MOD(MONTH(B2)-4, 12)/3 + 0.01, 0)
Weekday name            =TEXT(B2, "dddd")
Due date in 10 workdays =WORKDAY(B2, 10, Holidays)
Hours between times     =(End - Start) * 24
```

Level 3 · Logic & lookups

Logical functions

```
=IF(H2>=50000, "Key account", "Regular")
=IF(H2>=50000, "Gold", IF(H2>=20000, "Silver", "Bronze"))          nested IF
=IFS(H2>=50000, "Gold", H2>=20000, "Silver", TRUE, "Bronze")         IFS (365/2019+)
=SWITCH(D2, "N", "North", "S", "South", "Other")                      match exact values
=IF(AND(Status="Paid", Days<=30), "On time", "Late")
=IF(OR(City="Pune", City="Mumbai"), "Maharashtra", "Other")
=IFERROR(H2/G2, 0)          =IFNA(VLOOKUP(...), "Not found")         error handling
=ISBLANK(A2)  =ISNUMBER(SEARCH("pro", B2))  =ISERROR(...)             IS functions
```

Lookups: the evolution

Function	Formula (find a product's price)	Pros	Cons
VLOOKUP	`=VLOOKUP(A2, Products!$A:$C, 3, FALSE)`	Known everywhere	Only looks right; breaks if a column is inserted; FALSE is easy to forget
HLOOKUP	`=HLOOKUP(J2, $B$1:$M$2, 2, FALSE)`	Horizontal tables	Same limits
INDEX + MATCH	`=INDEX(Products!C:C, MATCH(A2, Products!A:A, 0))`	Any direction, robust, fast	Two functions to learn
Two-way INDEX/MATCH	`=INDEX(Grid, MATCH(rep,Reps,0), MATCH(month,Months,0))`	Row + column lookup	
XLOOKUP (365/2021)	`=XLOOKUP(A2, Products[ID], Products[Price], "Missing")`	Any direction, default not-found, last-match search, returns whole rows	Not in Excel 2016/2019
Approximate match	`=VLOOKUP(Score, Slabs, 2, TRUE)` or `=XLOOKUP(Score, Slab, Rate, , -1)`	Tax slabs, grade bands, commission tiers	Slab table must be sorted ascending
CHOOSE	`=CHOOSE(WEEKDAY(B2,2), "Mon","Tue",...)`	Pick from a list by position; enables left VLOOKUP	
INDIRECT / OFFSET	`=SUM(INDIRECT(A1&"!H:H"))`	Dynamic references	Volatile (slow); avoid in big files

Real example: commission slabs

Slab table: 0 → 2%, 1,00,000 → 3%, 5,00,000 → 5%. Rep sales ₹3,40,000. `=XLOOKUP(340000, Slab, Rate, , -1)` finds the largest slab ≤ sales → 3% → commission ₹10,200.

Level 4 · Data tools

Excel Tables (Ctrl+T): do this first, always

- Formulas fill down automatically and use names: `=[@Qty]*[@Price]`.
- New rows are included in pivots, charts and formulas automatically.
- Built-in Total Row, slicers and banded styles.

Cleaning toolkit

Remove Duplicates	Data → Remove Duplicates (choose key columns)
Text to Columns	Split by delimiter or fixed width; also converts text dates
Find & Replace	Wildcards * and ?; replace double spaces; Options → match entire cell
Go To Special → Blanks	Fill blanks with the value above: select, type `=` + up arrow, `Ctrl` + `Enter`
TRIM/CLEAN/PROPER	Fix spaces, non-printing characters, case
Data Validation	Drop-down lists (List source), whole number limits, dates, custom formula rules (`=COUNTIF(A:A,A2)=1` blocks duplicates), input messages and error alerts
Copilot "Clean data" (365)	Detects inconsistent text, spaces and number formats and suggests fixes

Sorting and filtering

Multi-level and custom-list sorts; filter by value, colour, number/date conditions; Advanced Filter with criteria ranges; `SUBTOTAL` / `AGGREGATE` for visible rows; the modern `=FILTER()` and `=SORT()` formulas return live results without touching the source.

Conditional formatting

Rules, data bars, colour scales, icon sets, and formula-based rules (highlight the row, overdue items, weekends, duplicates). Manage rules: Home → Conditional Formatting → Manage Rules to set order and "Stop If True".

Protection and collaboration

Unlock input cells (Format Cells → Protection → uncheck Locked), then Review → Protect Sheet. Protect Workbook locks structure. Co-authoring in OneDrive/SharePoint; comments with @mentions; Show Changes history.

Level 5 · Analysis

Pivot tables step by step

1. Click in a clean Table → Insert → PivotTable → New Worksheet.
2. Drag fields: Rows (City), Columns (Year), Values (Sum of Amount), Filters (Category).
3. Value Field Settings: Sum/Count/Average; **Show Values As** % of Grand Total, % of Row, Difference From previous (MoM/YoY), Running Total, Rank.
4. Group dates (Years, Quarters, Months) and numbers into bins.
5. Insert Slicers and a Timeline; connect them to several pivots via Report Connections.
6. Design: Tabular layout, repeat labels, subtotals on/off, styles. Refresh with `Alt` + `F5`; Refresh All `Ctrl` + `Alt` + `F5`.

- **Calculated field**: Margin % = Profit / Sales (works on sums).
- **GETPIVOTDATA**: pull a number from a pivot into a report cell that survives re-layout.
- **Pivot charts** follow the pivot's filters and slicers.
- Formula alternatives in 365: `=GROUPBY()` and `=PIVOTBY()` give pivot results that update automatically.

What-if analysis

Goal Seek	Data → What-If → Goal Seek: "What price gives ₹10 lakh profit?" Set cell = profit, To value = 1000000, By changing = price
Scenario Manager	Save Best / Base / Worst sets of inputs and produce a summary report
Data Table	One- or two-variable sensitivity grid: EMI for different rates × tenures
Solver (add-in)	Optimise with constraints: product mix that maximises profit within machine hours
Forecast Sheet	Data → Forecast Sheet: exponential-smoothing forecast with confidence bounds (`FORECAST.ETS`)

Real example: loan EMI

₹10 lakh, 9% annual, 5 years: `=PMT(9%/12, 60, -1000000)` = **₹20,758** per month. A two-variable Data Table shows EMI for 8–11% × 3–7 years in one grid.

Statistics in Excel

Enable the **Analysis ToolPak** (File → Options → Add-ins): Descriptive Statistics, Histogram, Correlation, Regression, t-Tests, ANOVA. Functions: CORREL, SLOPE, INTERCEPT, RSQ, FORECAST.LINEAR, T.TEST, CHISQ.TEST, NORM.DIST.

Level 6 · Modern Excel

Dynamic arrays (Excel 365 / 2021+)

One formula returns many values that "spill" into neighbouring cells. Refer to the whole spill with `#`: `=SUM(F2#)`. A #SPILL! error means something blocks the spill range.

Function	Example	Returns
UNIQUE	`=UNIQUE(Sales[City])`	Distinct cities
SORT / SORTBY	`=SORT(Sales, 5, -1)`	Table sorted by 5th column descending
FILTER	`=FILTER(Sales, (Sales[City]="Pune")*(Sales[Amount]>5000), "None")`	Rows meeting both conditions (* = AND, + = OR)
SEQUENCE	`=SEQUENCE(12,1,DATE(2026,4,1),31)`	Series of numbers/dates
XLOOKUP / XMATCH	`=XLOOKUP(A2, IDs, Sales[[Name]:[City]])`	Several columns at once
TAKE / DROP / CHOOSECOLS / CHOOSEROWS	`=TAKE(SORT(Sales,5,-1), 10)`	Top 10 rows
VSTACK / HSTACK	`=VSTACK(Jan!A2:E100, Feb!A2:E100)`	Append sheets
TEXTSPLIT / TEXTBEFORE / TEXTAFTER	`=TEXTSPLIT(A2, ",")`	Split text to columns
GROUPBY / PIVOTBY / PERCENTOF	`=GROUPBY(Sales[City], Sales[Amount], SUM, , , -2)`	Summary sorted by total
REGEXTEST / REGEXEXTRACT / REGEXREPLACE	`=REGEXEXTRACT(A2, "[A-Z]{5}[0-9]{4}[A-Z]")`	PAN-format code from text

LET and LAMBDA

```
=LET(sales, SUM(Sales[Amount]), cost, SUM(Sales[Cost]),
     margin, (sales - cost) / sales,
     TEXT(margin, "0.0%"))                                  name intermediate results

Define a reusable function: Formulas → Name Manager → New
```

```
Name:  GSTINCL     Refers to:  =LAMBDA(amount, rate, amount * (1 + rate))
Use:   =GSTINCL(H2, 18%)
```

Level 7 · Power tools

Power Query: repeatable data cleaning

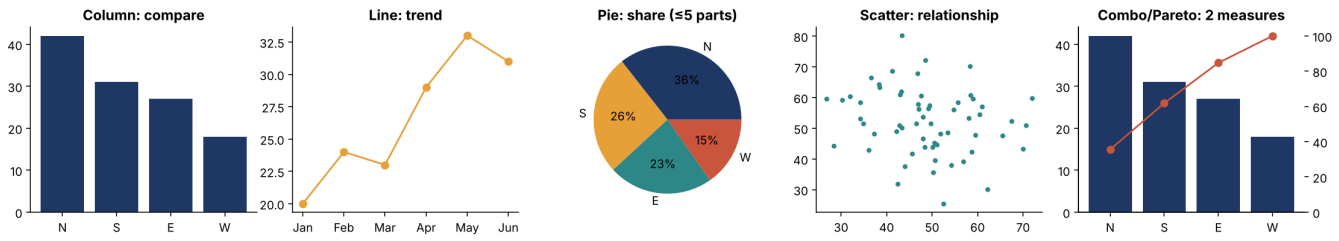
1. Data → Get Data → From File/Folder/Database/Web. "From Folder" combines every monthly file in a folder automatically.
2. In the editor: set types, remove columns, filter rows, split columns, replace values, trim, fill down, unpivot, merge (join) and append queries, group by, add conditional and custom columns.
3. Close & Load to a Table or only to the Data Model.
4. Next month: drop the new file in the folder → Refresh. Every step reruns.

> **Tip** Power Query records each click as an M-language step (Applied Steps). The same engine is in Power BI, so this skill transfers directly.

Power Pivot and the Data Model

- Load several tables (Sales, Products, Calendar) and connect them with relationships (Data → Relationships or the Diagram View), instead of VLOOKUPs.
- Write DAX measures: `Total Sales := SUM(Sales[Amount])`, `Sales LY := CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Calendar[Date]))`.
- Pivot tables built on the Data Model can handle millions of rows and use those measures.

Level 8 · Charts & dashboards



Start from the question, not the chart gallery.

Advanced chart	Use	How
Combo (column + line, secondary axis)	Revenue and margin % together	Change Chart Type → Combo
Pareto	80/20: few products, most complaints	Insert → Statistic → Pareto
Waterfall	Profit bridge: revenue → costs → profit	Insert → Waterfall; set totals
Histogram / Box & Whisker	Distributions	Insert → Statistic charts
Funnel	Visit → cart → checkout → paid	Insert → Funnel
Map chart	Sales by state	Insert → Maps (needs region names)
Sparklines	Mini trend inside a cell	Insert → Sparklines
Dynamic chart	Chart changes with a drop-down	Chart on a FILTER/INDEX result driven by a validation list

Dashboard design rules

✔ One screen; KPIs in the top row (cards with value + vs target + trend arrow)
✔ Main trend chart, then breakdowns; details last
✔ Slicers/timelines on one side, connected to all pivots
✔ Consistent colours: one highlight colour, greys for context
✔ Gridlines off, aligned shapes, titles that state the insight
✔ Raw data and calculations on hidden or separate sheets; protect the dashboard

The workbook's **18 Dashboard Case Study** sheet builds a monthly sales dashboard for a sample retailer with live KPIs, a store league table and charts.

Level 9 · Automation & AI

Macros and VBA

1. Show the Developer tab (File → Options → Customize Ribbon).
2. Record Macro → do the steps (format, sort, filter) → Stop. Use Relative References for steps that should work from the current cell.
3. Save as **.xlsm**. Run from View → Macros or assign to a button.
4. Edit in the VBA editor (`Alt` + `F11`):

```
Sub FormatReport()
    Dim ws As Worksheet
    For Each ws In ThisWorkbook.Worksheets          ' loop every sheet
        ws.Rows(1).Font.Bold = True
        ws.Columns("A:H").AutoFit
```

```
    Next ws
    MsgBox "Formatted " & ThisWorkbook.Worksheets.Count & " sheets"
End Sub
```

Office Scripts and Power Automate (cloud)

Excel for the web → Automate tab → Record Actions creates a TypeScript Office Script. Power Automate can run it on a schedule or when a file arrives: e.g. "every Monday 9 AM, refresh the sales workbook and email the summary to the team".

AI in Excel

Copilot chat	"Show sales by city as a chart", "Add a column for margin %", "What are the top 3 insights?"
Formula help	Generate, explain ("Explain this formula") and fix formulas from the error indicator
`=COPILOT()`	AI inside a cell: classify reviews, extract cities, summarise comments row by row
Agent Mode	Multi-step builds (a full model or dashboard) from a single request
Python in Excel `=PY()`	pandas, seaborn and scikit-learn inside the grid, run in Microsoft's cloud
External assistants	ChatGPT, Claude or Gemini to write formulas, VBA or Power Query M; always test on a copy

> Check AI-written formulas on a few rows you can verify by hand, and keep sensitive data out of tools your company has not approved.

Errors & auditing

Error	Cause	Fix
#N/A	Lookup value not found (extra spaces, text vs number)	TRIM, VALUE, IFNA/XLOOKUP not-found argument
#VALUE!	Wrong type (text in maths)	Clean the input; VALUE()
#DIV/0!	Divide by zero/blank	`=IFERROR(a/b,0)` or `=IF(b=0,"",a/b)`
#REF!	Referenced cell deleted	Undo; rebuild with INDEX/MATCH or Tables
#NAME?	Misspelt function or unquoted text; function not in your Excel version	Check spelling and version
#SPILL!	Dynamic array blocked	Clear the cells in the way
#NUM!	Impossible number (IRR not converging, SQRT of negative)	Check inputs

Error	Cause	Fix
Circular reference	Formula refers to itself	Formulas → Error Checking → Circular References

Auditing tools (Formulas tab): Trace Precedents/Dependents, Evaluate Formula (step through), Watch Window, Show Formulas (`Ctrl` + `` ` ``), Inquire add-in to compare workbooks.

Interview Q&A

Q VLOOKUP vs INDEX/MATCH vs XLOOKUP?

VLOOKUP looks only to the right and breaks when columns move. INDEX/MATCH looks any direction and is robust. XLOOKUP does both in one function with a built-in not-found value and is the modern default in Excel 365.

Q How would you find duplicates?

Conditional formatting → Duplicate Values; `=COUNTIF($A:$A,A2)>1`; or UNIQUE / Remove Duplicates to get a clean list.

Q What is a pivot cache and why does it matter?

A snapshot of the source data stored with the pivot; you must Refresh to see changes. Pivots built from the same source can share one cache.

Q When would you use Power Query instead of formulas?

When the same cleaning must be repeated on new files (monthly exports), when combining many files, or when the data is too messy for formulas. It is recorded once and refreshed.

Q How do you make a chart update automatically?

Base it on an Excel Table or a dynamic array formula; new rows are then included without editing the chart range.

Q What does SUMPRODUCT do?

Multiplies arrays element by element and sums the results, so it can do weighted sums and multi-condition counts without helper columns.

Q How do you protect formulas but let users enter data?

Unlock input cells, keep formula cells locked, then Protect Sheet (optionally with a password and allowed actions).

Resources

- ▶ YouTube: Simplilearn Excel full course · ▶ YouTube: Simplilearn advanced Excel tutorial · ▶ YouTube: Simplilearn Excel pivot table tutorial
- ▶ YouTube: Leila Gharani XLOOKUP · ▶ YouTube: Leila Gharani dynamic arrays FILTER SORT UNIQUE · ▶ YouTube: Leila Gharani Power Query
- ▶ YouTube: Chandoo Excel dashboard tutorial · ▶ YouTube: ExcelIsFun Excel basics to advanced · ▶ YouTube: MyOnlineTrainingHub Excel GROUPBY PIVOTBY
- ▶ YouTube: Kenji Excel copilot tutorial · ▶ YouTube: Excel VBA macros for beginners
- 📘 Microsoft Excel help and learning · 📘 Excel functions (alphabetical)
- 📘 Excel Campus: 2025 updates · 📘 Excel blog (What's new)